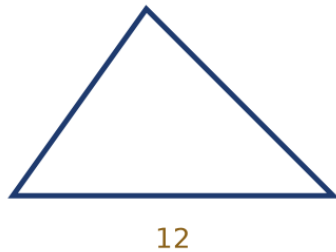


Geometry & Trigonometry

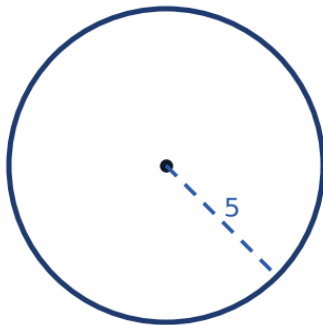


Area and perimeter of basic shapes

- 1 Find the area of a triangle with base 12 and height 7.

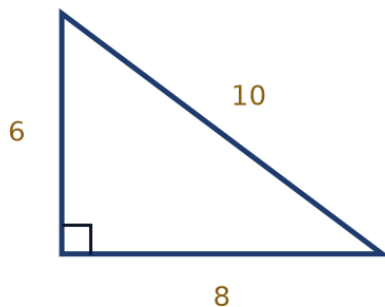


- 2 Find the area and circumference of a circle with radius 5, in terms of π .

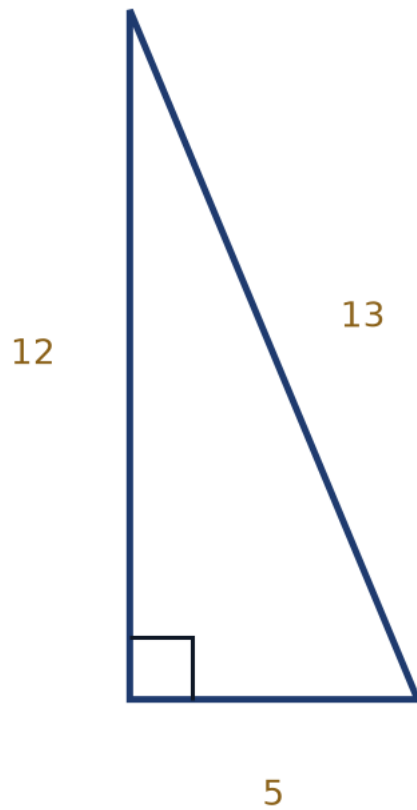


The Pythagorean theorem

- 3 A right triangle has legs 6 and 8. Find the length of the hypotenuse.

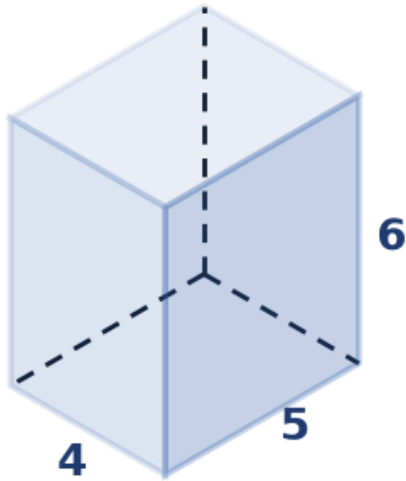


- 4 A ladder 13 feet long leans against a wall, with its base 5 feet from the wall. How high up the wall does the ladder reach?

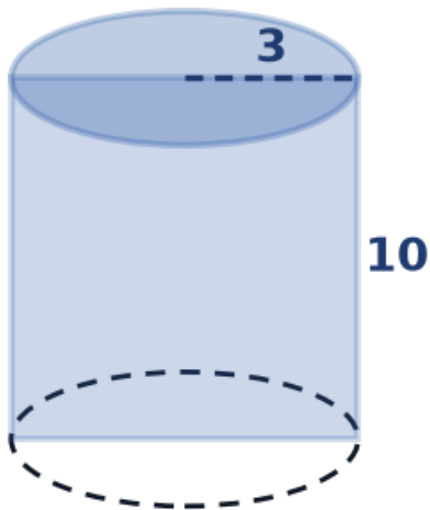


Volume and surface area of solids

- 5 Find the volume of a rectangular prism with dimensions 4 by 5 by 6.



- 6 Find the volume of a cylinder with radius 3 and height 10, in terms of π .

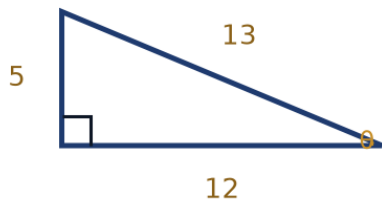


Similar triangles and proportional reasoning

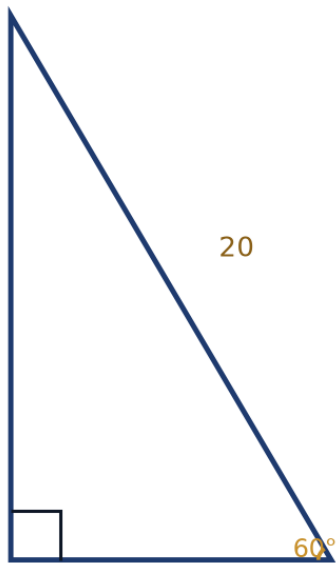
- 7 Triangle ABC is similar to triangle DEF , with $AB = 6$, $DE = 9$, and $BC = 8$. Find EF .
- 8 A tree casts a shadow 20 feet long, while a 6-foot person standing nearby casts a shadow 4 feet long. Find the height of the tree using similar triangles.
-

Right triangle trigonometry

- 9 In a right triangle, the angle θ has an opposite side of length 5 and hypotenuse 13. Find $\sin \theta$, $\cos \theta$, and $\tan \theta$.

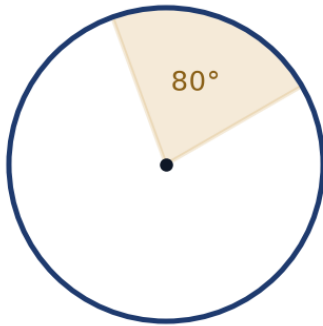


- 10 A 20-foot ladder leans against a wall, making a 60° angle with the ground. Find the height the ladder reaches on the wall, to the nearest tenth.



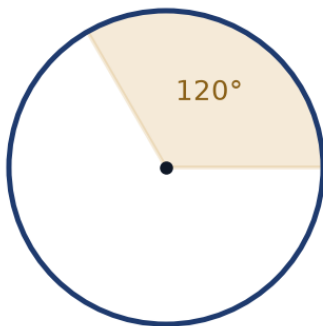
Angle relationships and circle theorems

- 11** Two parallel lines are cut by a transversal. If one angle measures 65° , find the measure of its corresponding angle and its co-interior (same-side interior) angle.
- 12** A central angle in a circle measures 80° . Find the measure of the inscribed angle that subtends the same arc.

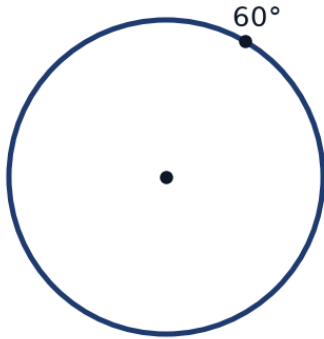


The unit circle and radian measure

- 13** Convert 120° to radians, in terms of π .

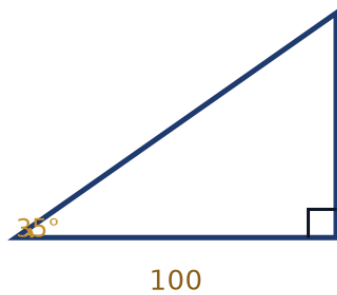


- 14 Find the exact value of $\cos\left(\frac{\pi}{3}\right)$ using the unit circle.



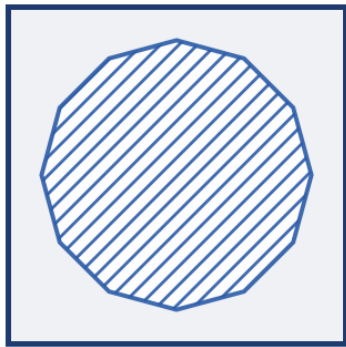
Applying geometry and trigonometry to real-world contexts

- 15 A surveyor stands 100 feet from the base of a building and measures the angle of elevation to the top as 35° . Find the height of the building, to the nearest foot.



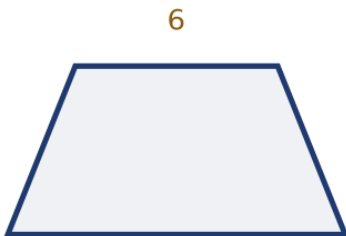
Composite area and shaded regions

- 16 A square with side length 10 has a circle of radius 4 cut out of its center. Find the remaining area, in terms of π .



10

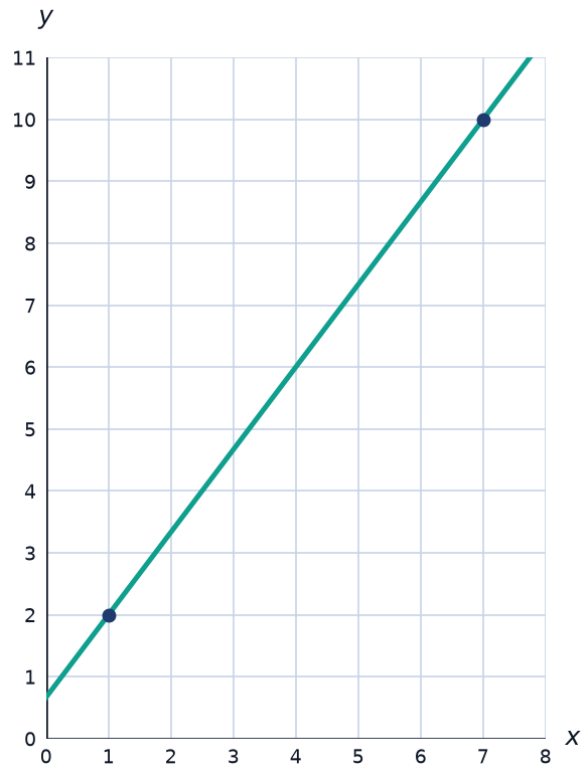
- 17 Find the area of a trapezoid with parallel sides 6 and 10, and height 5.



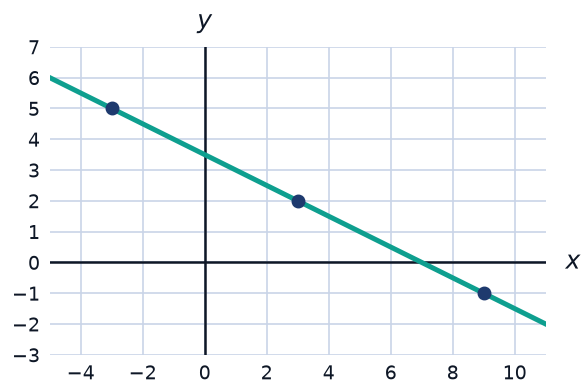
10

Coordinate geometry: distance and midpoint

- 18 Find the distance between the points $(1, 2)$ and $(7, 10)$.

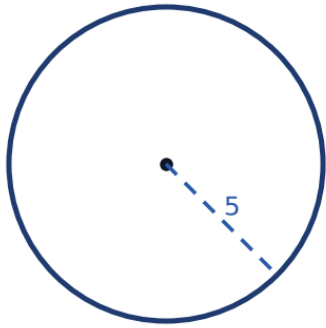


- 19 Find the midpoint of the segment connecting $(-3, 5)$ and $(9, -1)$.

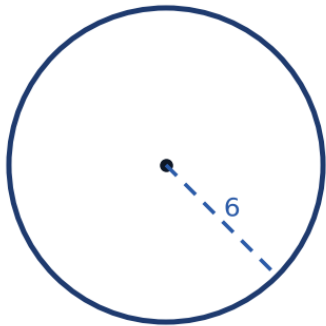


Equation of a circle

- 20 Write the equation of a circle with center $(2, -3)$ and radius 5.

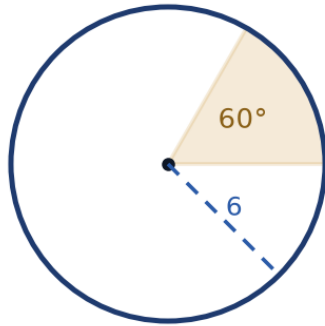


- 21 Find the center and radius of the circle $(x + 1)^2 + (y - 4)^2 = 36$.

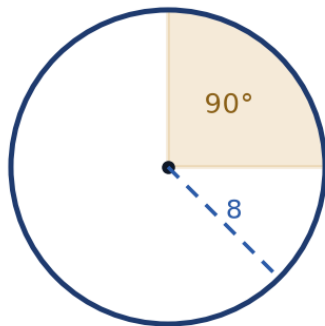


Arc length and sector area

- 22 A circle has radius 6. Find the arc length of a 60° sector, in terms of π .



- 23 Find the area of a 90° sector of a circle with radius 8, in terms of π .



Multiple choice — easy

- 24 A right triangle has legs 20 and 48. Find the length of the hypotenuse.

- a. 53
- b. 68
- c. 52
- d. 51

- 25 A right triangle has legs 12 and 16. Find the length of the hypotenuse.
- a. 28
 - b. 19
 - c. 21
 - d. 20
- 26 A right triangle has legs 18 and 80. Find the length of the hypotenuse.
- a. 81
 - b. 83
 - c. 98
 - d. 82
- 27 A right triangle has legs 15 and 36. Find the length of the hypotenuse.
- a. 40
 - b. 51
 - c. 38
 - d. 39
- 28 A right triangle has legs 10 and 24. Find the length of the hypotenuse.
- a. 26
 - b. 25
 - c. 27
 - d. 34

- 29 A right triangle has legs 9 and 12. Find the length of the hypotenuse.
- a. 14
 - b. 21
 - c. 15
 - d. 16
- 30 A right triangle has legs 24 and 32. Find the length of the hypotenuse.
- a. 40
 - b. 39
 - c. 41
 - d. 56
- 31 A right triangle has legs 27 and 120. Find the length of the hypotenuse.
- a. 123
 - b. 124
 - c. 122
 - d. 147
- 32 A right triangle has legs 36 and 160. Find the length of the hypotenuse.
- a. 164
 - b. 165
 - c. 163
 - d. 196

- 33 A right triangle has legs 21 and 72. Find the length of the hypotenuse.
- a. 76
 - b. 74
 - c. 75
 - d. 93
- 34 A right triangle has legs 24 and 45. Find the length of the hypotenuse.
- a. 50
 - b. 52
 - c. 51
 - d. 69
- 35 A right triangle has legs 40 and 42. Find the length of the hypotenuse.
- a. 57
 - b. 59
 - c. 82
 - d. 58
- 36 A right triangle has legs 8 and 15. Find the length of the hypotenuse.
- a. 23
 - b. 17
 - c. 18
 - d. 16

- 37 A right triangle has legs 6 and 8. Find the length of the hypotenuse.
- a. 14
 - b. 11
 - c. 9
 - d. 10
- 38 A right triangle has legs 14 and 48. Find the length of the hypotenuse.
- a. 51
 - b. 49
 - c. 50
 - d. 62
-

Multiple choice — medium

- 39 Find the distance between the points $(-5, -1)$ and $(-35, -73)$.
- a. 79
 - b. 102
 - c. 77
 - d. 78
- 40 Find the distance between the points $(-3, -8)$ and $(2, -20)$.
- a. 12
 - b. 17
 - c. 13
 - d. 14

- 41 Find the distance between the points $(3, -3)$ and $(-12, -39)$.
- a. 51
 - b. 38
 - c. 40
 - d. 39
- 42 Find the distance between the points $(-8, 5)$ and $(-14, 13)$.
- a. 9
 - b. 10
 - c. 14
 - d. 11
- 43 Find the distance between the points $(1, -1)$ and $(25, -33)$.
- a. 39
 - b. 41
 - c. 40
 - d. 56
- 44 Find the distance between the points $(7, -4)$ and $(28, -76)$.
- a. 76
 - b. 74
 - c. 75
 - d. 93

- 45 Find the distance between the points $(-2, 6)$ and $(-11, -34)$.
- a. 41
 - b. 40
 - c. 49
 - d. 42
- 46 Find the distance between the points $(1, -1)$ and $(-8, 39)$.
- a. 41
 - b. 42
 - c. 49
 - d. 40
- 47 Find the distance between the points $(-6, 8)$ and $(-46, -34)$.
- a. 58
 - b. 57
 - c. 82
 - d. 59
- 48 Find the distance between the points $(3, -6)$ and $(15, -22)$.
- a. 20
 - b. 19
 - c. 21
 - d. 28

- 49 Find the distance between the points $(-6, -4)$ and $(12, 20)$.
- a. 31
 - b. 30
 - c. 29
 - d. 42
- 50 Find the distance between the points $(7, -3)$ and $(10, -7)$.
- a. 6
 - b. 7
 - c. 5
 - d. 4
- 51 Find the distance between the points $(-4, -3)$ and $(20, 42)$.
- a. 51
 - b. 69
 - c. 50
 - d. 52
- 52 Find the distance between the points $(-7, 7)$ and $(9, 37)$.
- a. 34
 - b. 33
 - c. 46
 - d. 35

53 Find the distance between the points $(8, 0)$ and $(-12, -21)$.

- a. 28
- b. 41
- c. 29
- d. 30

Multiple choice — hard

54 In a right triangle, the angle θ has an opposite side of length 7, adjacent side 24, and hypotenuse 25. Find $\cos \theta$.

- a. $\frac{24}{7}$
- b. $\frac{24}{25}$
- c. $\frac{25}{24}$
- d. $\frac{25}{7}$

55 In a right triangle, the angle θ has an opposite side of length 12, adjacent side 16, and hypotenuse 20. Find $\tan \theta$.

- a. $\frac{4}{3}$
- b. $\frac{5}{3}$
- c. $\frac{3}{4}$
- d. $\frac{5}{4}$

- 56 In a right triangle, the angle θ has an opposite side of length 6, adjacent side 8, and hypotenuse 10. Find $\sin \theta$.
- a. $\frac{5}{3}$
 - b. $\frac{3}{5}$
 - c. $\frac{4}{3}$
 - d. $\frac{5}{4}$
- 57 In a right triangle, the angle θ has an opposite side of length 7, adjacent side 24, and hypotenuse 25. Find $\tan \theta$.
- a. $\frac{25}{7}$
 - b. $\frac{25}{24}$
 - c. $\frac{7}{24}$
 - d. $\frac{24}{7}$
- 58 In a right triangle, the angle θ has an opposite side of length 3, adjacent side 4, and hypotenuse 5. Find $\tan \theta$.
- a. $\frac{5}{3}$
 - b. $\frac{4}{3}$
 - c. $\frac{5}{4}$
 - d. $\frac{3}{4}$
- 59 A circle has radius 5. Find the sector area of a 180° sector, in terms of π .
- a. $\frac{23}{2}\pi$
 - b. 25π
 - c. $\frac{27}{2}\pi$
 - d. $\frac{25}{2}\pi$

60 A circle has radius 11. Find the arc length of a 180° sector, in terms of π .

- a. 10π
- b. 22π
- c. 11π
- d. 12π

61 A circle has radius 3. Find the arc length of a 45° sector, in terms of π .

- a. $\frac{1}{10}\pi$
- b. $\frac{3}{4}\pi$
- c. $\frac{3}{2}\pi$
- d. $\frac{7}{4}\pi$

62 A circle has radius 5. Find the sector area of a 270° sector, in terms of π .

- a. $\frac{75}{2}\pi$
- b. $\frac{71}{4}\pi$
- c. $\frac{75}{4}\pi$
- d. $\frac{79}{4}\pi$

63 A circle has radius 7. Find the sector area of a 135° sector, in terms of π .

- a. $\frac{147}{8}\pi$
- b. $\frac{139}{8}\pi$
- c. $\frac{147}{4}\pi$
- d. $\frac{155}{8}\pi$

- 64 Triangle ABC is similar to triangle DEF , with $AB = 35$, $DE = 14$, and $BC = 35$. Find EF .
- a. 16
 - b. 12
 - c. 35
 - d. 14
- 65 Triangle ABC is similar to triangle DEF , with $AB = 12$, $DE = 18$, and $BC = 8$. Find EF .
- a. 14
 - b. 8
 - c. 10
 - d. 12
- 66 Triangle ABC is similar to triangle DEF , with $AB = 32$, $DE = 24$, and $BC = 12$. Find EF .
- a. 9
 - b. 7
 - c. 11
 - d. 12
- 67 Triangle ABC is similar to triangle DEF , with $AB = 21$, $DE = 7$, and $BC = 15$. Find EF .
- a. 5
 - b. 15
 - c. 7
 - d. 3

68 Triangle ABC is similar to triangle DEF , with $AB = 20$, $DE = 12$, and $BC = 15$. Find EF .

a. 15

b. 11

c. 7

d. 9